INCH-POUND

MIL-F-48277B (AR)
AMENDMENT 4
22 February 1989
SUPERSEDING
AMENDMENT 3
14 June 1983

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M739
LESS BOOSTER PELLET

This Amendment forms a part of Military Specification MIL-F-48277B (AR), dated 24 August, 1978, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.5: Delete in its entirety and substitute the following:

"3.5 Fuze leak test. The Fuze Assemblies, less Booster Pellet and Booster Cup, shall comply with MIL-STD-331, Test No. 108 except that the duration of immersion shall be five (5 plus or minus 0.5) minutes and the pressure shall be three (3 plus or minus 0.5) PSI gauge."

PAGE 5

4.3.1: Add subparagraph f. as follows:

"f. Fifty (50) each (Dwg. 9258606) Fuze Body Assemblies for the Leak Test (see 4.5.8)."

PAGE 6

* 4.4.2.a Add the following:

"Requests for switching from "Normal" to "Reduced" inspection shall satisfy the requirements of MIL-STD-105 and shall be submitted to the government for approval. Such requests shall include supportive inspection history data. Prior to such submission the contractor's in-process inspection procedures shall have been approved by AMSMC-QAR-Q (D)."

PAGE 8

After Page 8, add new 4.4.2.2.1 as shown herein. Page (8a).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48277B (AR)
AMENDMENT 4

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2.1	Pin, Firing	1	1	9333937
				NEXT HIGHER ASSEMBLY 9294606

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of shaft and profile of point		0.40%	3.1	Gage
102	Unit weight, max.		0.40%	3.1	Balance
103	Pull test	80		3.2	4.5.3
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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PAGE 10

4.4.2.4: Add the following:

"Major 102; Shrinkage defects, unfilled or incomplete component, Sample Size (a), Visual inspection.

Major 103; Presence of flash or gates, Sample Size (a), Visual inspection.

Minor 203; Cavity identification missing or improper, Sample Size (a), Visual inspection.

Notes: (a) This inspection applies only when molded, plastic component is produced. For qualification of a mold, inspect three (3) consecutive samples from each mold cavity. During normal production, inspect one (1) sample each from beginning and end of each production run (see 4.5.30)."

PAGE 11

* 4.4.2.5: For Major 101, delete "80" and add "0.40% AQL".

PAGE 26

* 4.4.2.20: For Major 101, delete "80" and add "0.40% AQL".

PAGE 27

4.4.2.21: Change Drawing Number 9258649 to Drawing Number 9343014.

PAGE 53

4.4.2.43: Delete Critical 1 in its entirety. For Critical 3, change "Gage" to "4.5.14/Gage."

* 4.4.2.43: For Major 104, delete "80" and add "0.40% AQL".

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PAGE 56

4.4.2.46: Delete Critical 1 in its entirety. For Critical 2, change "Gage" to "4.5.15/Gage."

Add Special Defect as follows:

"Special.

a: Presence of spinlock springs, 100%; 3.2, Gage (prior to non-arming test)."

* 4.4.2.46: For Major 101, delete "80" and add "0.40% AQL".

PAGE 59

4.4.2.49: Delete Major 139 in its entirety.

PAGE 60

4.4.2.49: Add Minor 204 as follows:

"204. Length from base of fuze to base of wrench slot flange, min., 0.65%, 3.1, Gage."

PAGE 64

4.4.2.53: Change the sample size for Major 103, Leak Test, from 80 to 12.

PAGE 68

4.4.3.3.1: Add the following Note:

"Note: The total ballistic sample will be one hundred fifty (150) fuzes, since the sample of fifteen (15) from 4.4.3.2.1 are included in the proving ground tests."

PAGE 69

4.4.3.6: Add new paragraph as follows:

"4.4.3.6 Fuze leak test.

4.4.3.6.1 Initial production. Beginning with the first lot produced and continuing until three consecutive lots have been found acceptable, a sample of fifty (50) fuzes shall be randomly selected from each lot for use in the leak test specified in 4.5.8. If three (3) or more assemblies are classed defective, the lot shall be rejected.

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4.4.3.6.2 Normal production. After three consecutive lots have complied with the criteria of 4.4.3.6.1, the sample size shall be as specified in 4.4.2.53. If two (2) or more assemblies are classed defective, the lot shall be rejected. If only one (1) assembly is classed defective, a second sample of thirty-eight (38) assemblies shall be tested and the lot rejected if two or more additional failures are found."

PAGE 70

Delete 4.5.3 in its entirety and substitute the following:

"4.5.3 Firing pin/assembly pull test. The firing pin (Dwg. 9333937) or firing pin assembly (Dwg. 9258618), whichever is applicable, shall be assembled to an applicable test fixture and an axial pull load of 80 pounds shall be applied to the firing pin shaft. If two or more firing pin shafts, during the testing of firing pin assemblies (9258618), come loose or separate, the lot of firing pin assemblies shall be rejected. If two or more firing pins, during the testing of firing pins (9333937), show evidence of deformation, i.e., the final over-all length exceeds the drawing requirement, the lot of firing pins shall be rejected. Any item subjected to the pull test shall not be returned to the lot."

PAGE 71

4.5.8: Delete in its entirety and substitute the following:

"4.5.8 Leak test. The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 108, except as provided in 3.5 and herein. When immersing the test assemblies in the water solution, care should be taken to displace all externally entrapped air. The test assemblies should be submerged in a horizontal position for testing. After completion of the test and drying of the samples, the assemblies shall be sectioned to reveal all cavities internal to the various seals. Examine all external and internal surfaces of parts for evidence of water entry using ultra-violet light. Any assembly that does not comply with the requirements of 3.5 shall be classed defective. Assemblies subjected to this test shall not be returned to the lot."

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Add 4.5.8.1 as follows:

"4.5.8.1 Alternate leak test. The following optional method may be used to perform the waterproofness test required by 3.5. This method requires pressurizing the test assemblies in air prior to submersion in water and does not require the use of any additive to the water. In an approved test apparatus, pressurize the test assemblies in air at 3 ± 0.5 psig for 5 ± 0.5 minutes. Then lower the assemblies into water while maintaining the required air pressure above the water. The assemblies, in a horizontal orientation, shall be agitated while submerged to remove surface-entrapped air. After the assemblies have been agitated and there is a minimum of 3 inches of water above the assemblies (remaining in horizontal orientation), release the air pressure above the water and observe the test assemblies for the presence of bubbles for a minimum of 30 seconds. Any assembly with a continuous stream of bubbles in excess of one bubble per second shall be classed defective. Assemblies subjected to this test shall not be returned to the lot. Assemblies which pass this test may also be used to satisfy the sample requirements for Jolt and Jumble testing (see 4.4.3.1) or for Propagation testing (see 4.4.3.5)."

* Delete 4.5.12 and add the following:

"4.5.12 Lead assembly push-out test. The safing and arming module subassembly shall be disassembled and assembled to the applicable test fixture and the required load shall be applied to the lead cup assembly from a direction opposite the green end. Sampling shall be conducted in accordance with MIL-STD-105D, special inspection level S-3 at the specified AQL. If one or more lead cup assemblies or displaced material do not remain flush or below flush with the bottom plate the lot shall be rejected. A dummy Lead may be used for test purposes. Assemblies subjected to this test shall not be returned to the lot. Note: This test may be performed on safing and arming module assemblies (9294833) after completion of the push test on the setback pin disk provided the subassembly has not been damaged by the test (see 4.5.19)."

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4.5.14: Delete the last sentence in the paragraph and substitute the following:

"Any assembly which is not in the full safe position (rotor locked by both spinlocks) after being spun at 1100 ± 50 RPM's for a minimum of 1.5 seconds shall be classed defective and removed from the lot."

4.5.15: Delete the sentence beginning "Any assembly that does not remain ..." and substitute the following sentence:

"Any assembly which is not in the safe, out-of-line position after being spun at 4,000 RPM's minimum for a minimum of 0.5 seconds shall be classed defective and removed from the lot."

* Delete 4.5.16 and add the following:

"4.5.16 Push test on pallet shaft. The lower plate and shaft assembly after staking or pressing shall be held in an appropriate test fixture and the required push-out force shall be applied axially to the pallet shaft from the staked or press-fit side of the plate. Sampling shall be conducted in accordance with MIL-STD-105D, special inspection level S-3, at the specified AQL. If one or more assemblies fail to withstand the two-pound push-out force, the lot shall be rejected. Assemblies subjected to this test may be returned to the lot."

* Delete 4.5.19 and add the following:

"4.5.19 Push test on setback pin disk. The safing and arming module assemblies shall be disassembled and assembled to the applicable test fixture and the required load shall be applied through the setback pin to the setback pin disk. Sampling shall be conducted in accordance with MIL-STD-105D, special inspection level S-3, at the specified AQL. If one or more setback pin disks do not remain flush or below flush with the bottom plate under the applied load, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. Note: This test may be followed by a lead assembly push-out test. (see 4.5.12).

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PAGE 75

* Delete 4.5.21 and add the following:

"4.5.21 Firing pin and housing assembly load test. The firing pin and housing assembly shall be assembled to an applicable test fixture and the required load shall be applied to an axial direction at a uniform rate. When the applied load reaches 100 pounds, the maximum deflection shall not exceed .035 inches. Sampling shall be conducted in accordance with MIL-STD-105D, special inspection level S-3 at the specified AQL. If one or more assemblies exceed the maximum deflection of .035 inches, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot."

PAGE 78

Add 4.5.27.3 as follows:

"4.5.27.3 Definition of dud. Fuzes which fail to function where intended shall be classified as duds. For SQ firings into a plywood target at 500 feet, intended functioning of the fuze is on impact with the target (on the target). For Delay Mode (D) firings into a plywood target at 500 feet, intended functioning of the fuze is behind the target and before subsequent impact with the ground."

PAGE 79

Add 4.5.30 as follows:

"4.5.30 Production verification of mold. The first component from each cavity produced during a production run and the last component from each cavity produced during the same production run shall be examined in accordance with 4.4.2. A production run is defined as the continuous forming of consecutive components by a mold or die which has been prepared in accordance with its operating specifications, and maintained in the same operating condition between the first and last samples which represent that production run. The production run represented by these two samples shall be rejected if either of the two samples are found to be defective. Samples shall be permitted to cool to room temperature (80 + 20°F) prior to examination for compliance with dimensional requirements.

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The contractor may elect to cut off certain quantities of components during a production run. This is acceptable, provided one component is selected for inspection at the end of that quantity. This sample may also be considered as the first component produced during the next quantity of the same production run. The rejection criteria remain the same."

PAGE 80

Add 6.8 as follows:

"6.8 International standardization agreements. Certain provisions (4.4.2.53, 4.4.2.54, 4.4.3.1, 4.4.3.2, 4.4.3.3, 4.4.3.5, 4.5.8) of this specification are the subject of international standardization agreement (QSTAG 626). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required."

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